192425192

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EXP21:

```
#include <stdio.h>

#include <string.h>

#include <stdint.h>


static const int
IP[64]={58,50,42,34,26,18,10,2,60,52,44,36,28,20,12,4,62,54,46,38,30,22,14,6,64,56,48,40,32,24,16,8,57,49,41,33,25,17,9,1,59,51,43,35,27,19,11,3,61,53,45,37,29,21,13,5,63,55,47,39,31,23,15,7};

static const int
FP[64]={40,8,48,16,56,24,64,32,39,7,47,15,55,23,63,31,38,6,46,14,54,22,62,30,37,5,45,13,53,21,61,29,36,4,44,12,52,20,60,28,35,3,43,11,51,19,59,27,34,2,42,10,50,18,58,26,33,1,41,9,49,17,57,25};

static const int
EX[48]={32,1,2,3,4,5,4,5,6,7,8,9,8,9,10,11,12,13,12,13,14,15,16,17,16,17,18,19,20,21,20,21,22,23,24,25,24,25,26,27,28,29,28,29,30,31,32,1};

static const int
PP[32]={16,7,20,21,29,12,28,17,1,15,23,26,5,18,31,10,2,8,24,14,32,27,3,9,19,13,30,6,22,11,4,25};

static const int
PC1[56]={57,49,41,33,25,17,9,1,58,50,42,34,26,18,10,2,59,51,43,35,27,19,11,3,60,52,44,36,63,55,47,39,31,23,15,7,62,54,46,38,30,22,14,6,61,53,45,37,29,21,13,5,28,20,12,4};

static const int
PC2[48]={14,17,11,24,1,5,3,28,15,6,21,10,23,19,12,4,26,8,16,7,27,20,13,2,41,52,31,37,47,55,30,40,51,45,33,48,44,49,39,56,34,53,46,42,50,36,29,32};

static const int SHIFTS[16]={1,1,2,2,2,2,2,2,1,2,2,2,2,2,2,1};

static const int SB[8][64]={

{14,4,13,1,2,15,11,8,3,10,6,12,5,9,0,7,0,15,7,4,14,2,13,1,10,6,12,11,9,5,3,8,4,1,14,8,13,6,2,11,15,12,9,7,3,10,5,0,15,12,8,2,4,9,1,7,5,11,3,14,10,0,6,13},

{15,1,8,14,6,11,3,4,9,7,2,13,12,0,5,10,3,13,4,7,15,2,8,14,12,0,1,10,6,9,11,5,0,14,7,11,10,4,13,1,5,8,12,6,9,3,2,15,13,8,10,1,3,15,4,2,11,6,7,12,0,5,14,9},

{10,0,9,14,6,3,15,5,1,13,12,7,11,4,2,8,13,7,0,9,3,4,6,10,2,8,5,14,12,11,15,1,13,6,4,9,8,15,3,0,11,1,2,12,5,10,14,7,1,10,13,0,6,9,8,7,4,15,14,3,11,5,2,12},
```

```
{7,13,14,3,0,6,9,10,1,2,8,5,11,12,4,15,13,8,11,5,6,15,0,3,4,7,2,12,1,10,14,9,10,6,9,0,12,11,7,
13,15,1,3,14,5,2,8,4,3,15,0,6,10,1,13,8,9,4,5,11,12,7,2,14},

{2,12,4,1,7,10,11,6,8,5,3,15,13,0,14,9,14,11,2,12,4,7,13,1,5,0,15,10,3,9,8,6,4,2,1,11,10,13,7,
8,15,9,12,5,6,3,0,14,11,8,12,7,1,14,2,13,6,15,0,9,10,4,5,3},

{12,1,10,15,9,2,6,8,0,13,3,4,14,7,5,11,10,15,4,2,7,12,9,5,6,1,13,14,0,11,3,8,9,14,15,5,2,8,12,
3,7,0,4,10,1,13,11,6,4,3,2,12,9,5,15,10,11,14,1,7,6,0,8,13},

{4,11,2,14,15,0,8,13,3,12,9,7,5,10,6,1,13,0,11,7,4,9,1,10,14,3,5,12,2,15,8,6,1,4,11,13,12,3,7,
14,10,15,6,8,0,5,9,2,6,11,13,8,1,4,10,7,9,5,0,15,14,2,3,12},

{13,2,8,4,6,15,11,1,10,9,3,14,5,0,12,7,1,15,13,8,10,3,7,4,12,5,6,11,0,14,9,2,7,11,4,1,9,12,14,
2,0,6,10,13,15,3,5,8,2,1,14,7,4,10,8,13,15,12,9,0,3,5,6,11}};


static void permute(const int *t,int n,const unsigned char *in,unsigned char *out){
    int i; for(i=0;i<n;i++) out[i]=in[t[i]-1];
}
static void to_bits(const unsigned char *b,int n,unsigned char *bits){
    int i; for(i=0;i<n*8;i++) bits[i]=(b[i/8]>>(7-i%8))&1;
}
static void from_bits(const unsigned char *bits,int n,unsigned char *b){
    int i; memset(b,0,n); for(i=0;i<n*8;i++) b[i/8]|=bits[i]<<(7-i%8);
}
static void des_ksched(const unsigned char key[8],unsigned char sk[16][48]){
    unsigned char kb[64],cd[56],c[28],d[28],t[28];
    int r,i,s;
    to_bits(key,8,kb); permute(PC1,56,kb,cd);
    memcpy(c,cd,28); memcpy(d,cd+28,28);
    for(r=0;r<16;r++){
        for(s=0;s<SHIFTS[r];s++){
            memcpy(t,c,28); for(i=0;i<28;i++) c[i]=t[(i+1)%28];
            memcpy(t,d,28); for(i=0;i<28;i++) d[i]=t[(i+1)%28];
        }
        memcpy(cd,c,28); memcpy(cd+28,d,28); permute(PC2,48,cd,sk[r]);
    }
```

```
}
static void des_crypt(const unsigned char in[8],unsigned char out[8],unsigned char sk[16]
[48],int dec){
    unsigned char b[64],ip[64],L[32],R[32],e[48],f[32],s[32],pre[64],ob[64];
    int r,i,j,row,col,v; unsigned char tmp;
    to_bits(in,8,b); permute(IP,64,b,ip);
    memcpy(L,ip,32); memcpy(R,ip+32,32);
    for(r=0;r<16;r++){
        const unsigned char *k=sk[dec?15-r:r];
        permute(EX,48,R,e); for(i=0;i<48;i++) e[i]^=k[i];
        for(j=0;j<8;j++){
            row=(e[6*j]<<1)|e[6*j+5];
            col=(e[6*j+1]<<3)|(e[6*j+2]<<2)|(e[6*j+3]<<1)|e[6*j+4];
            v=SB[j][row*16+col];
            for(i=0;i<4;i++) s[4*j+i]=(v>>(3-i))&1;
        }
        permute(PP,32,s,f);
        for(i=0;i<32;i++){ tmp=R[i]; R[i]=L[i]^f[i]; L[i]=tmp; }
    }
    memcpy(pre,R,32); memcpy(pre+32,L,32);
    permute(FP,64,pre,ob); from_bits(ob,8,out);
}
/* convenience: one 8-byte block with an 8-byte key */
static void des_block(const unsigned char in[8],unsigned char out[8],const unsigned char
key[8],int dec){
    unsigned char sk[16][48]; des_ksched(key,sk); des_crypt(in,out,sk,dec);
}
static void phex(const char *label,const unsigned char *b,int n){
    int i; printf("%s",label); for(i=0;i<n;i++) printf("%02X",b[i]); printf("\n");
}
```

```
/* Padding: a 1 bit (0x80) followed by as few 0 bits as needed.
   A padding block is added even when the message already fills the last block,
   so that the receiver can ALWAYS find and strip the padding unambiguously
   (otherwise a real message ending in 0x80 00.. would be mistaken for padding). */
static int pad(const unsigned char *in, int len, unsigned char *out) {
    int n = (len / 8 + 1) * 8;              /* always at least 1 byte of padding */
    memcpy(out, in, len); out[len] = 0x80; memset(out + len + 1, 0, n - len - 1);
    return n;
}
static int unpad(const unsigned char *in, int n) {
    int i = n - 1;
    while (i >= 0 && in[i] == 0) i--;
    return (i >= 0 && in[i] == 0x80) ? i : n;
}
static void xor8(unsigned char *a, const unsigned char *b) { int i; for (i = 0; i < 8; i++) a[i] ^= b[i]; }
static void show(const char *l, const unsigned char *b, int n) { int i; printf("%s", l); for (i = 0; i < n; i++) printf("%02X", b[i]); printf("\n"); }

int main() {
    unsigned char key[8] = {0x13,0x34,0x57,0x79,0x9B,0xBC,0xDF,0xF1};
    unsigned char iv[8]  = {0,1,2,3,4,5,6,7};
    char line[100];

    while (1) {
        unsigned char p[128], c[128], d[128], prev[8], x[8], ks[8];
        int len, n, i, back;
        printf("\nEnter message (empty line to stop): ");
        if (!fgets(line, sizeof line, stdin)) break;
        line[strcspn(line, "\n")] = 0;
        len = strlen(line);
```

```
        if (len == 0) break;


        n = pad((unsigned char *)line, len, p);
        printf("Message length = %d bytes, padded length = %d bytes (%d block(s))\n", len, n, n / 8);
        show("Padded   : ", p, n);


        for (i = 0; i < n; i += 8) des_block(p + i, c + i, key, 0);          /* ECB */
        show("ECB      : ", c, n);
        for (i = 0; i < n; i += 8) des_block(c + i, d + i, key, 1);
        back = unpad(d, n); d[back] = 0; printf("ECB back : %s\n", d);


        memcpy(prev, iv, 8);                                    /* CBC */
        for (i = 0; i < n; i += 8) { memcpy(x, p + i, 8); xor8(x, prev); des_block(x, c + i, key, 0); memcpy(prev, c + i, 8); }
        show("CBC      : ", c, n);
        memcpy(prev, iv, 8);
        for (i = 0; i < n; i += 8) { des_block(c + i, x, key, 1); xor8(x, prev); memcpy(d + i, x, 8); memcpy(prev, c + i, 8); }
        back = unpad(d, n); d[back] = 0; printf("CBC back : %s\n", d);


        memcpy(prev, iv, 8);                                    /* CFB (64-bit segments) */
        for (i = 0; i < n; i += 8) { des_block(prev, ks, key, 0); memcpy(x, p + i, 8); xor8(x, ks); memcpy(c + i, x, 8); memcpy(prev, x, 8); }
        show("CFB      : ", c, n);
        memcpy(prev, iv, 8);
        for (i = 0; i < n; i += 8) { des_block(prev, ks, key, 0); memcpy(x, c + i, 8); memcpy(prev, x, 8); xor8(x, ks); memcpy(d + i, x, 8); }
        back = unpad(d, n); d[back] = 0; printf("CFB back : %s\n", d);
    }
    printf("\nA padding block is always added so the receiver can always remove it correctly.\n");
```

```
    return 0;
}
```

OUTPUT:

```
Enter message (empty line to stop): HELLO
Message length = 5 bytes, padded length = 8 bytes (1 block(s))
Padded   : 48454C4C4F800000
ECB      : 9381DB78FC9EE756
ECB back : HELLO
CBC      : EF48A603A11225FD
CBC back : HELLO
CFB      : 96251085BF0F676F
CFB back : HELLO

Enter message (empty line to stop): ABCDEFGH
Message length = 8 bytes, padded length = 16 bytes (2 block(s))
Padded   : 41424344454647488000000000000000
ECB      : 0EE11BD2808EF0A187AB78D11E188DF6
ECB back : ABCDEFGH
CBC      : 28B7F6B79D80399958B4B4A44E32B99A
CBC back : ABCDEFGH
CFB      : 9F221F8DB5C92027CAB5AD9301E823C8
CFB back : ABCDEFGH

Enter message (empty line to stop):

A padding block is always added so the receiver can always remove it correctly.
```